

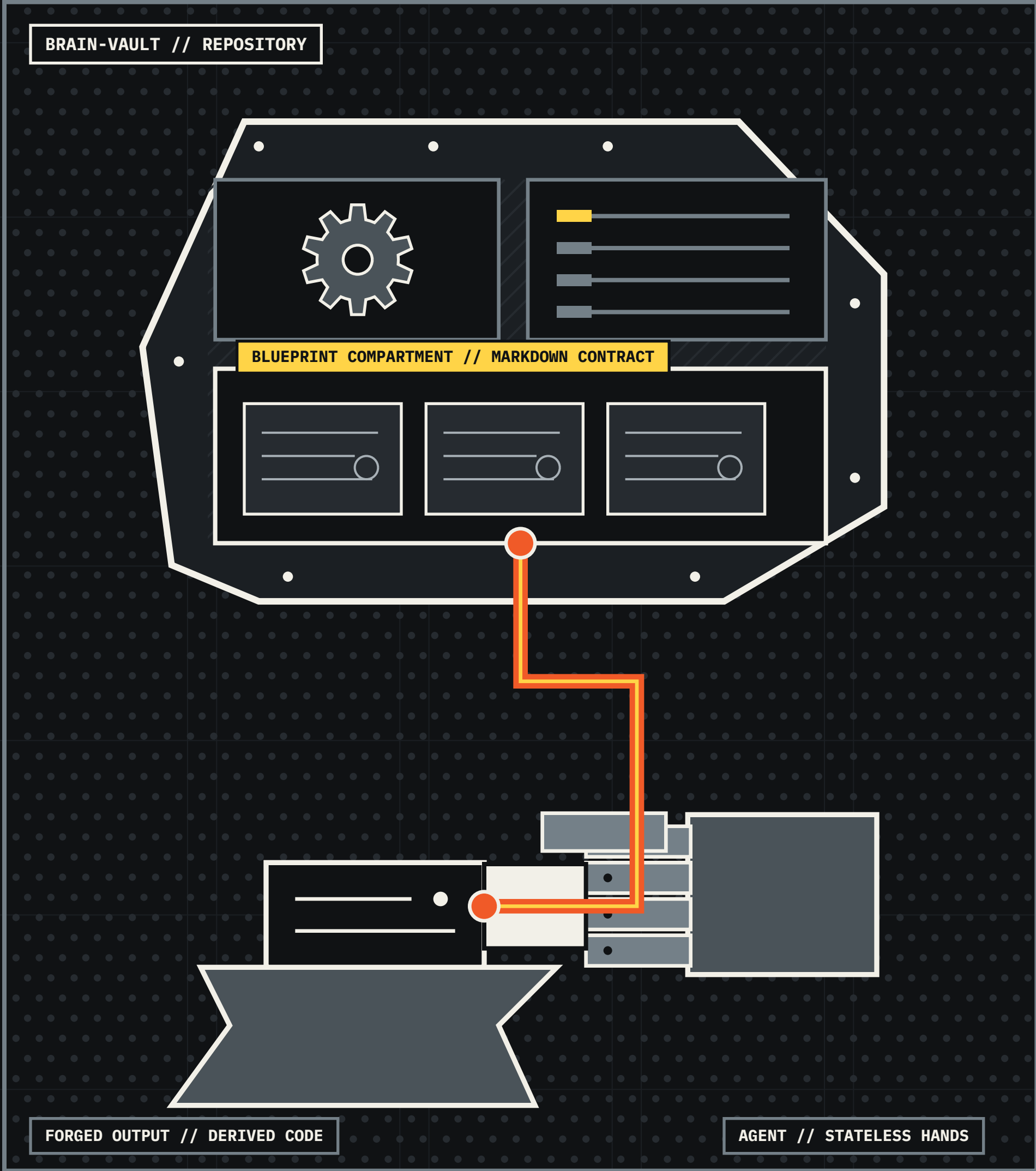
THE REPO
IS THE BRAIN.

THE AGENT
IS THE HANDS.

MARKDOWN IS
THE CONTRACT.

A workflow designed to outlive every agent session.

HUMAN IDEA → APPROVED SPEC → STATELESS AGENT → COMMITTED HANDOFF → VERIFIED BUILD



CHAT MEMORY IS A TRAP.

Prompts evaporate. Decisions drift. Work gets rediscovered.

PANEL 01 // EPHEMERAL PROMPT

CHAT-AS-SPEC → LOST CONTEXT

Prompts evaporate into thin air as soon as the window closes.

PANEL 02 // UNGOVERNED OUTPUT

UNTRACED CODE → SPEC DRIFT

Orphaned code blocks fall off an unmarked conveyor with no spec ID.

PANEL 03 // SESSION TERMINATION

UNWRITTEN HANDOFF → AMNESIA

An empty workstation faces a dead session; the next agent starts blind.

Move intelligence into the repository.

PERSISTENT FOUNDATION // VERSIONED SPECS • IMMUTABLE ADRs • WRITTEN HANDOFFS

REPO SURVIVES EVERY SESSION

THE SYSTEM, IN ONE LOOP

IDEA → APPROVED SPEC → PLAN / TASK → BUILD / TEST → HANDOFF
→ PUSH / PR → CI → HUMAN MERGE → DEPLOYED VERIFY

HUMAN LANE

AUTHORITY & GATES



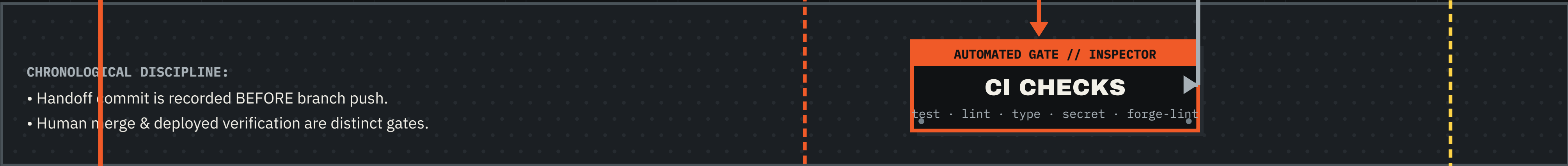
AGENT LANE

STATELESS EXECUTION



CI LANE

AUTOMATED INSPECTOR





RETURN BELT // .forge/STATE.md + .forge/LEDGER.md → NEXT STATELESS SESSION

The next agent resumes from committed state.



THE REPO IS THE MACHINE ROOM.

AGENTS.md

— one entry point

.forge/

— state, ledger, trace

vision/ · brainstorm/ · knowledge/

— intent and context

specs/ · decisions/

— contract and decisions

plans/ · tasks/

— execution

quality/ · ops/ · .github/

— controls

src/ · tests/

— derived product

CODE IS OUTPUT, NOT AUTHORITY

AGENTS.md



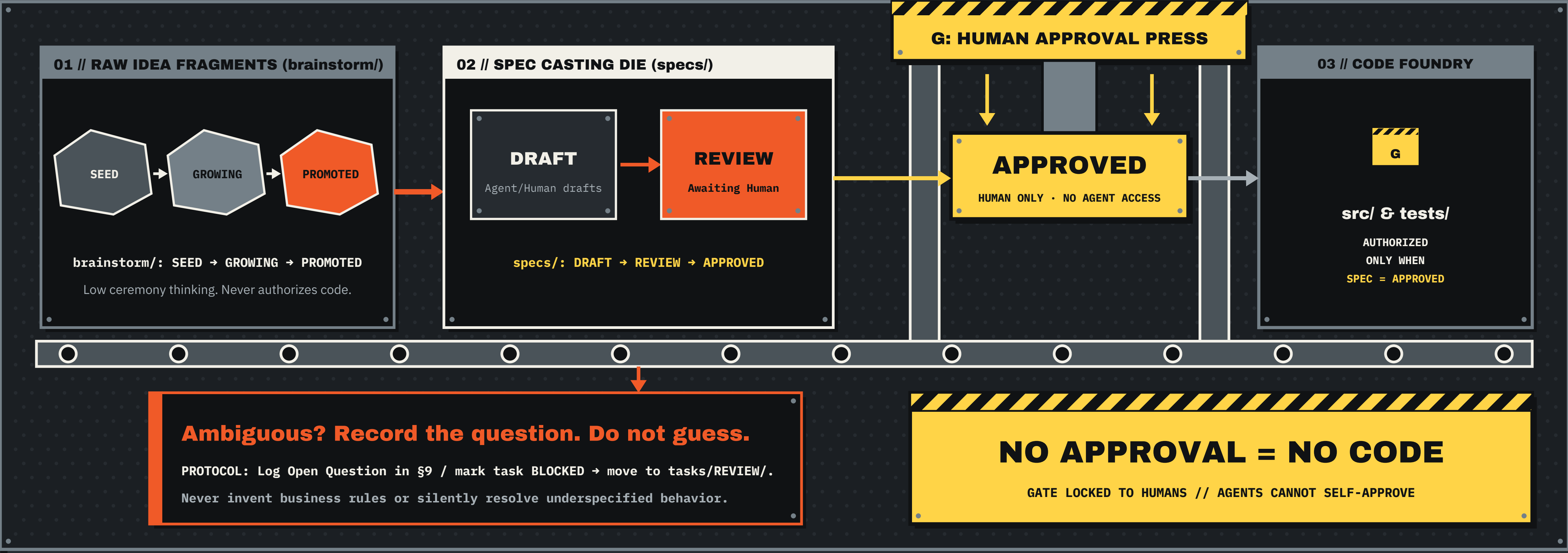
SINGLE FRONT ENTRY DOOR



IDEAS DO NOT SHIP.

brainstorm/: SEED → GROWING → PROMOTED

specs/: DRAFT → REVIEW → APPROVED



Only an **APPROVED** spec authorizes product code.

HARD RULE P1 & P7 // IF NO APPROVED SPEC EXISTS, DRAFT ONE IN specs/ AND STOP FOR HUMAN APPROVAL.

A SPEC IS A TESTABLE CONTRACT.

SPEC-012 / Password reset

STATUS: APPROVED

§5.2 BEHAVIORAL CONTRACT RULES (NUMBERED, DETERMINISTIC):

R1: 32 random bytes; store the SHA-256 token hash.

R2: 30-minute TTL; expired → 410.

R3: Single-use; replay → 410.

§6 ACCEPTANCE CRITERION (GIVEN / WHEN / THEN):

AC-2: Expired token → 410; password unchanged.

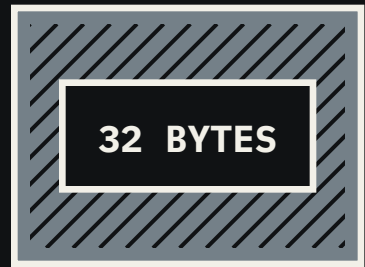
§10 EXECUTABLE VERIFICATION TARGET (tests/auth/test_reset.py):

TEST: test_ac_2_expired_token_410

Unresolved behavior → question and ADR before build.

0Q-1: Should reset invalidate sessions? → Record the question and an ADR before build.

FIG 06.A // EXPLODED LOCK MECHANISM (SPEC-012)



SHA-256 STORAGE CHAMBER

R1: Store token_hash UNIQUE
Raw token never persisted in DB

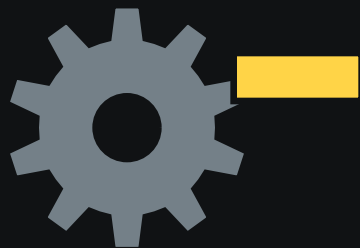


30-MINUTE TTL TIMER (R2)

IF now() > expires_at:

RETURN HTTP 410 GONE

ASSERT: user.password_hash UNCHANGED



ONE-USE RATCHET LATCH (R3)

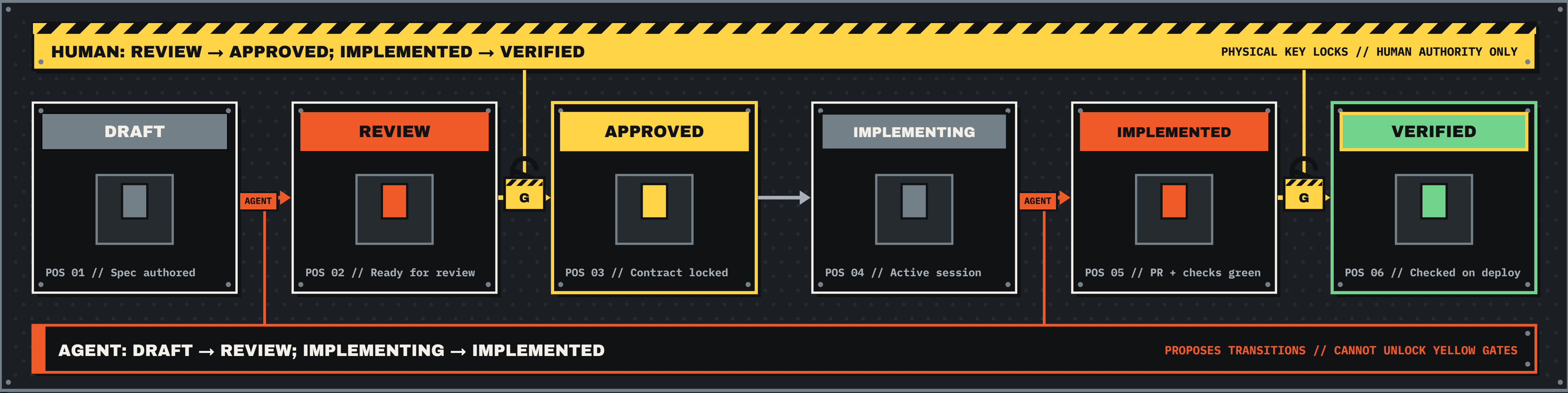
used_at != NULL → REPLAY BLOCKED

SECOND CONFIRM → HTTP 410 GONE

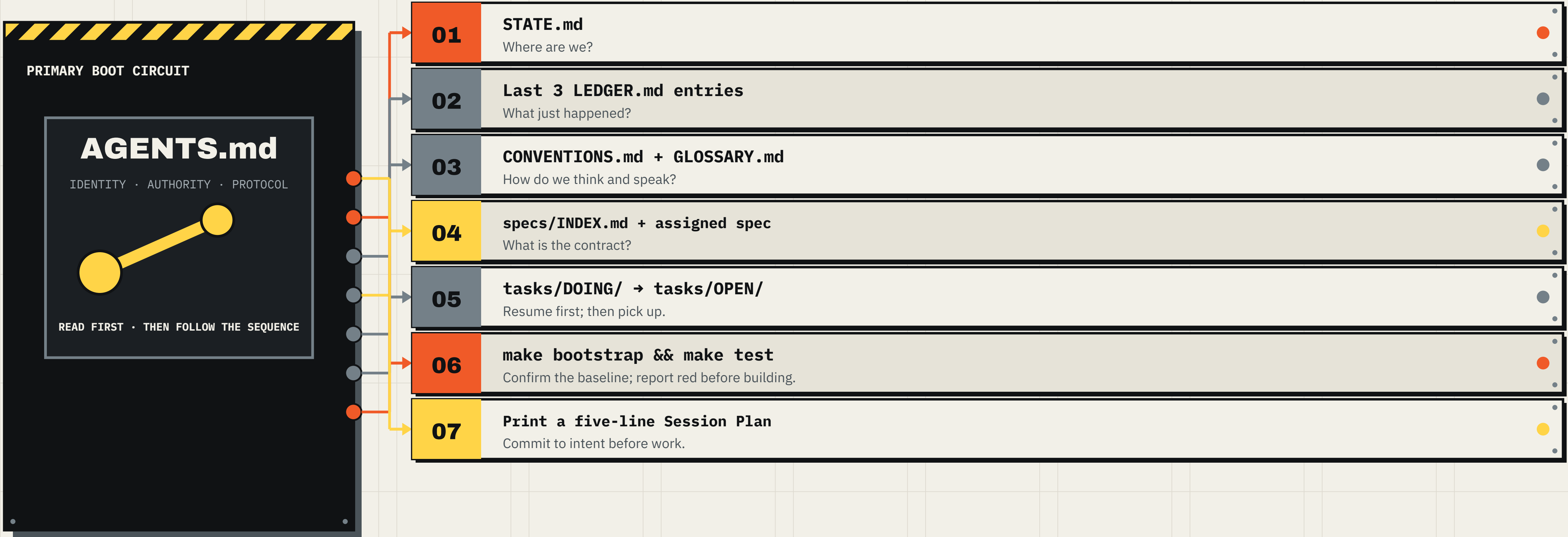
RULE BECOMES EXECUTABLE TEST

R2 (30m Timer) → AC-2 → test_ac_2_expired_token_410

STATUS IS THE CONTROL PLANE.



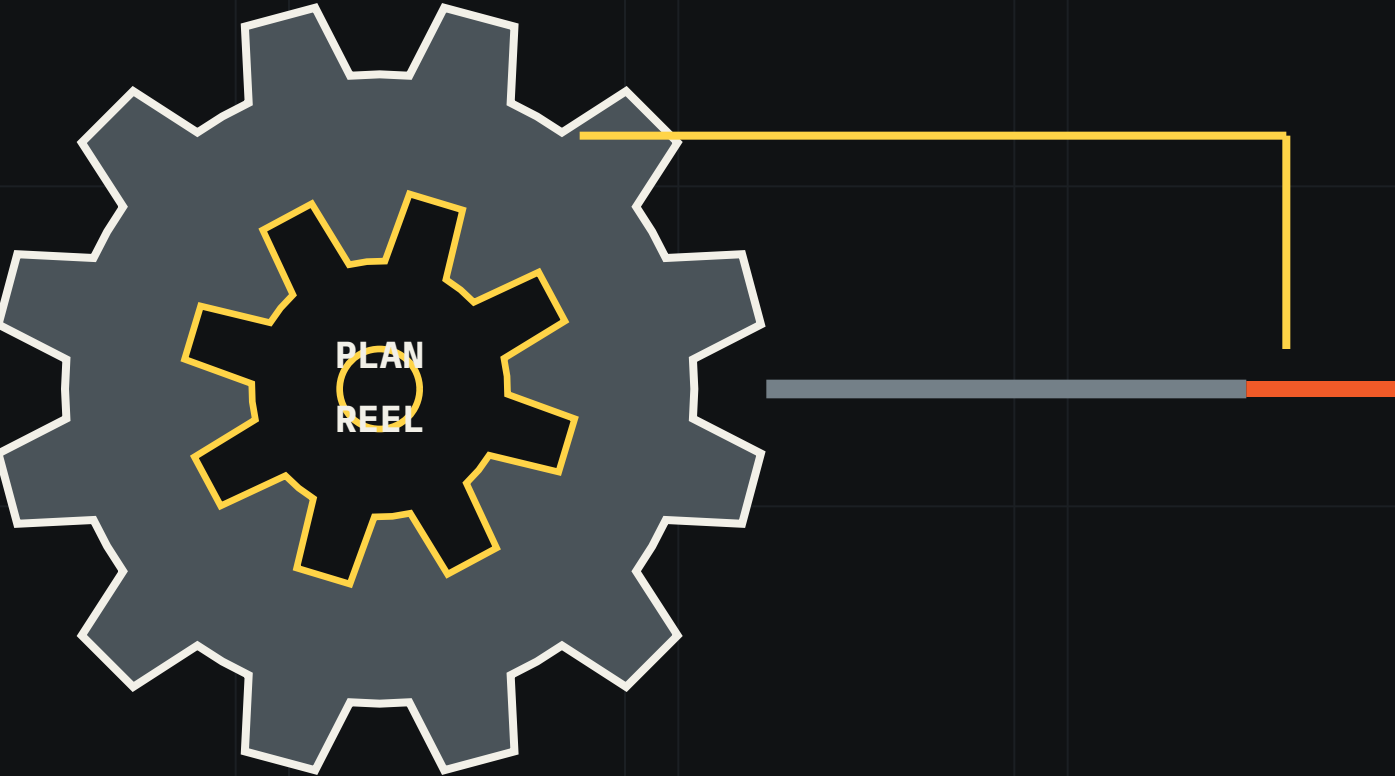
ONE FILE REBOOTS THE AGENT.



START WITH REPO STATE, NOT CHAT MEMORY.

NO BUILD UNTIL BASELINE IS UNDERSTOOD

PLANS TURN SPECS INTO WORK ORDERS.



WORK ORDER // AUTHORIZED FROM PLAN-004

- **T-032** **Reset confirm expiry and single-use**
-
-
-
-
-
-

ACCEPTANCE CRITERIA
AC-2 • AC-3

BRANCH
feat/T-032-reset-confirm

VERIFICATION – RUN; REPORT ACTUAL RESULT

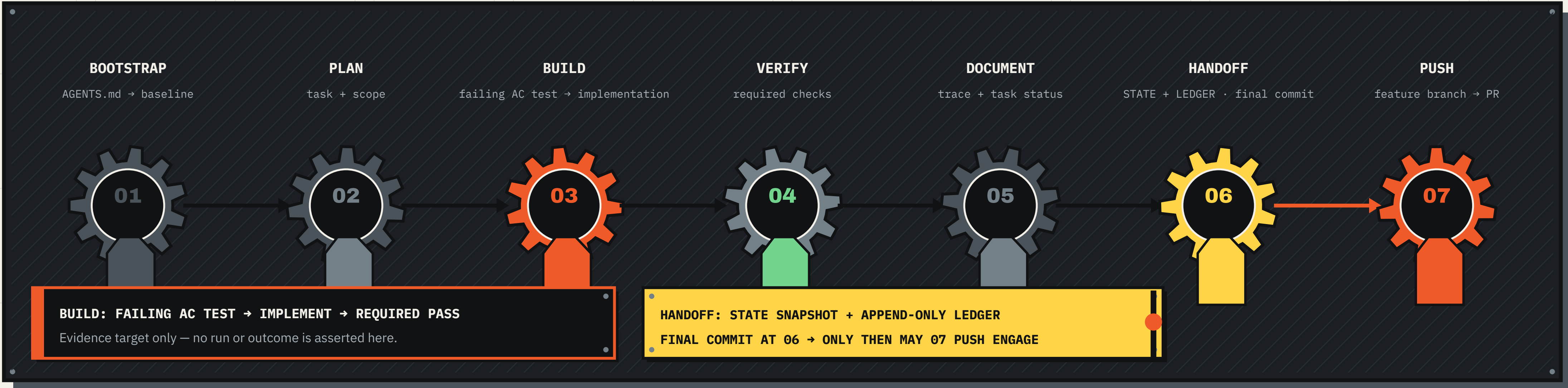
```
pytest tests/auth/test_reset.py -k "ac_2 or ac_3" -v
```

OUT OF SCOPE
Rate limiting (T-033)

1 PR

One task = one mergeable unit.

ONE SESSION. SEVEN MOVES.



01 BOOTSTRAP → 02 PLAN → 03 BUILD → 04 VERIFY
→ 05 DOCUMENT → 06 HANDOFF → 07 PUSH

PUSH: feature branch, then PR.

DONE MEANS EXIT 0.

REQUIRED PROCEDURE // NO RUN CLAIMED

01 // TEST FIRST

FAILING TEST

AC-2 → test_ac_2_expired_token_410

EVIDENCE SLOT – FILL FROM ACTUAL RUN
command · exit code · summary

TEST TARGET ONLY – THIS DIAGRAM IS NOT A TEST RESULT.

02 // IMPLEMENT + VERIFY

IMPLEMENT

FAILING TEST → IMPLEMENT → PASSING TEST

REQUIRED PASS – VERIFY BY RUNNING THE TEST

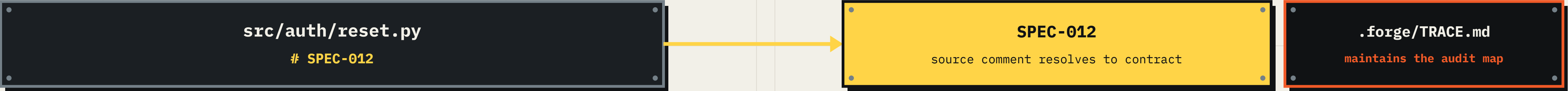
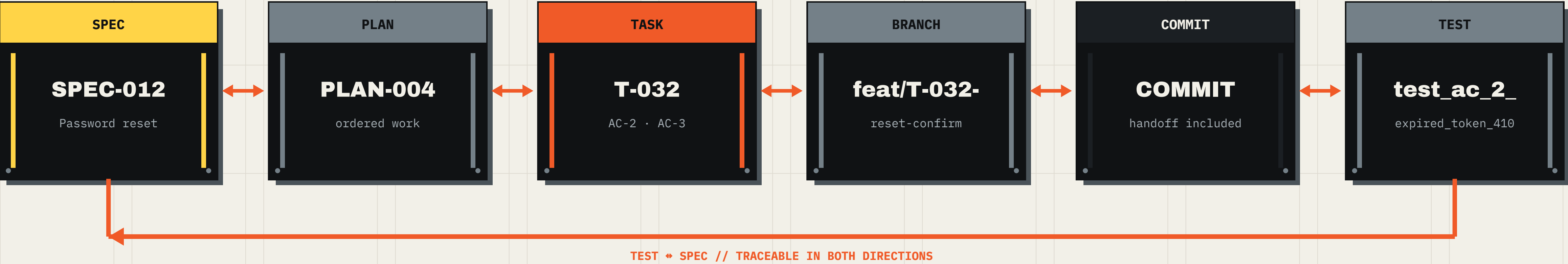
TARGETED RUN · FULL SUITE · LINT · TYPECHECK · SECRET SCAN
Actual command + exit code + summary → PR

NO FAKE TERMINAL LINES · NO INVENTED PASS COUNTS.

Paste the actual command, exit code, and summary into the PR.

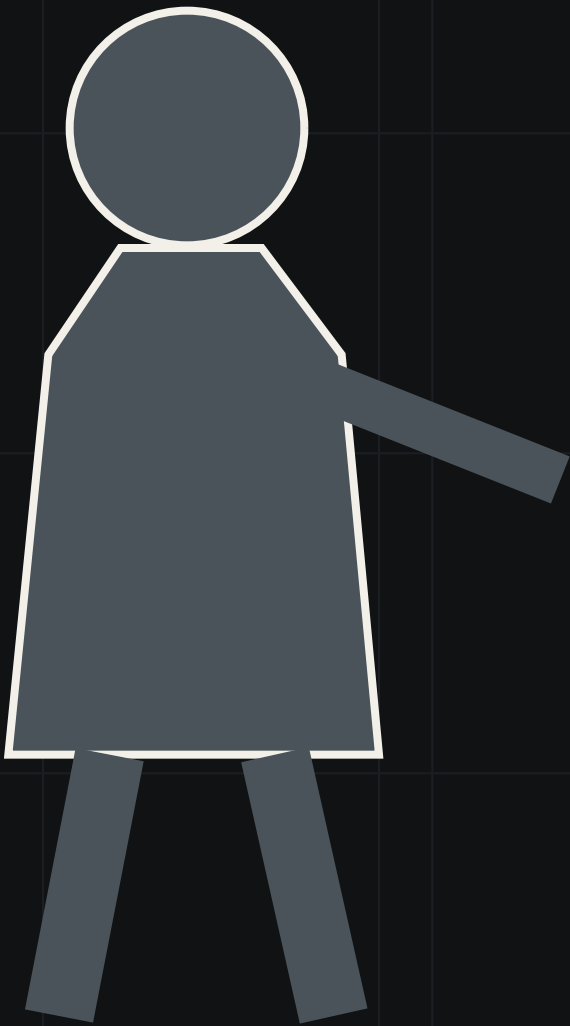
“Looks good” is not evidence.

EVERY LINK WALKS BOTH WAYS.



SPEC-012 ↔ PLAN-004 ↔ T-032 ↔ feat/T-032-reset-confirm
↔ COMMIT ↔ test_ac_2_expired_token_410 ↔ SPEC-012

THE HANDOFF IS THE MEMORY.



AGENT A // SESSION ENDS

REPOSITORY CASSETTE // COMMITTED CONTINUITY

STATE.md	current snapshot	OVERWRITE
LEDGER.md	session history	APPEND ONLY
TRACE.md	changed links	UPDATE MAP
tasks/	true status folder	MOVE FILES



AGENT B // READS RECORD

LAST COMMIT // HANDOFF SEALED

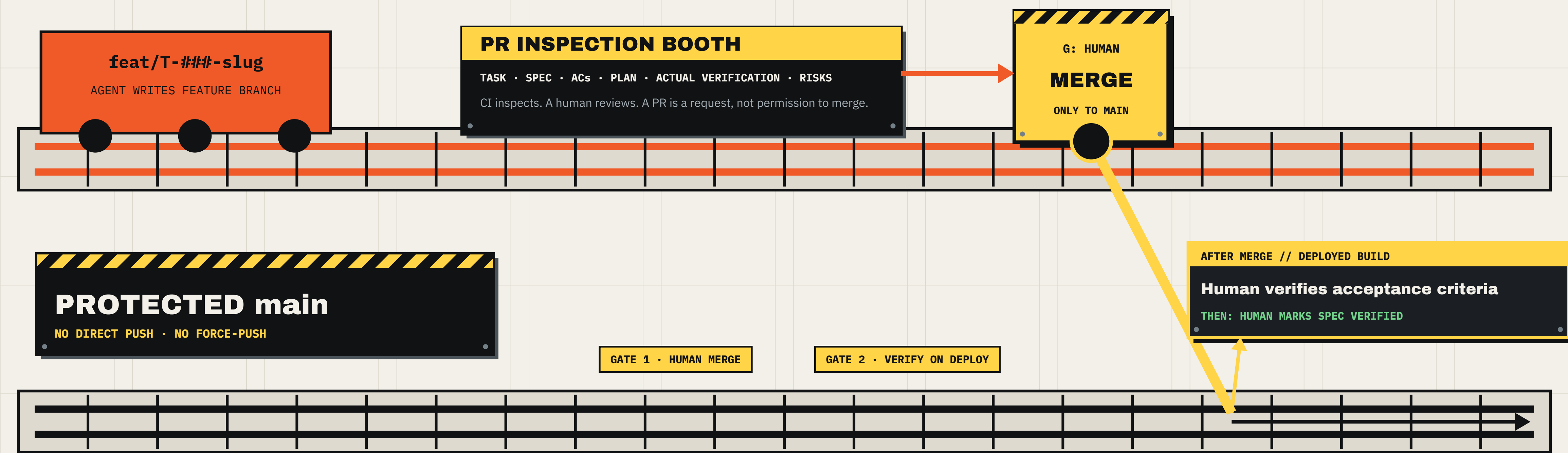
STATE + LEDGER + TRACE + TASK STATUS

PUSH BRANCH

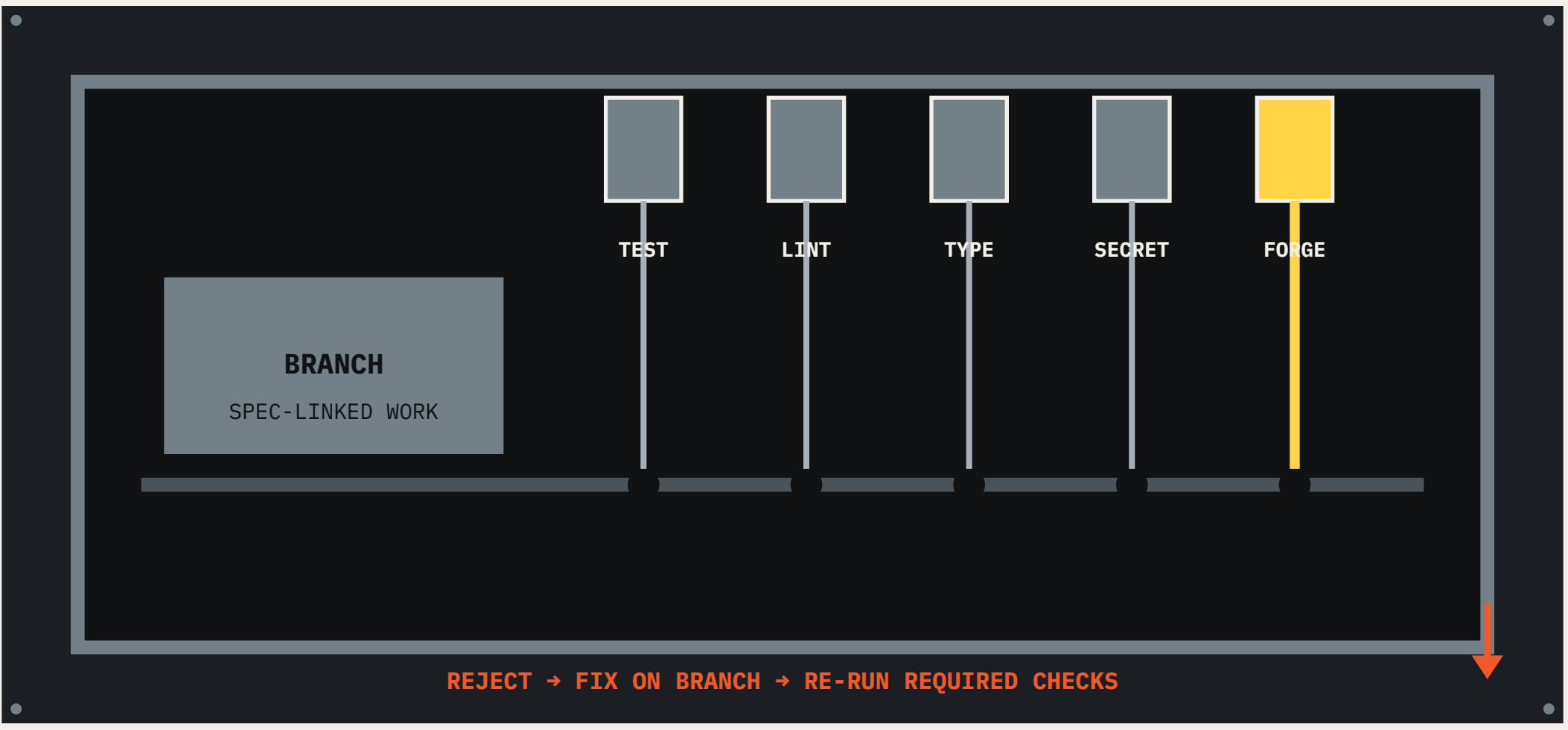
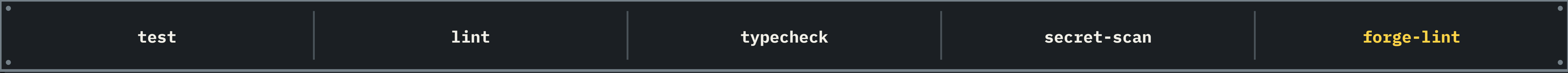
Commit handoff LAST → push branch.

NEXT: read STATE + latest LEDGER entries.

MAIN NEVER BELONGS TO THE AGENT.



CI IS AN INSPECTOR, NOT A CHEERLEADER.



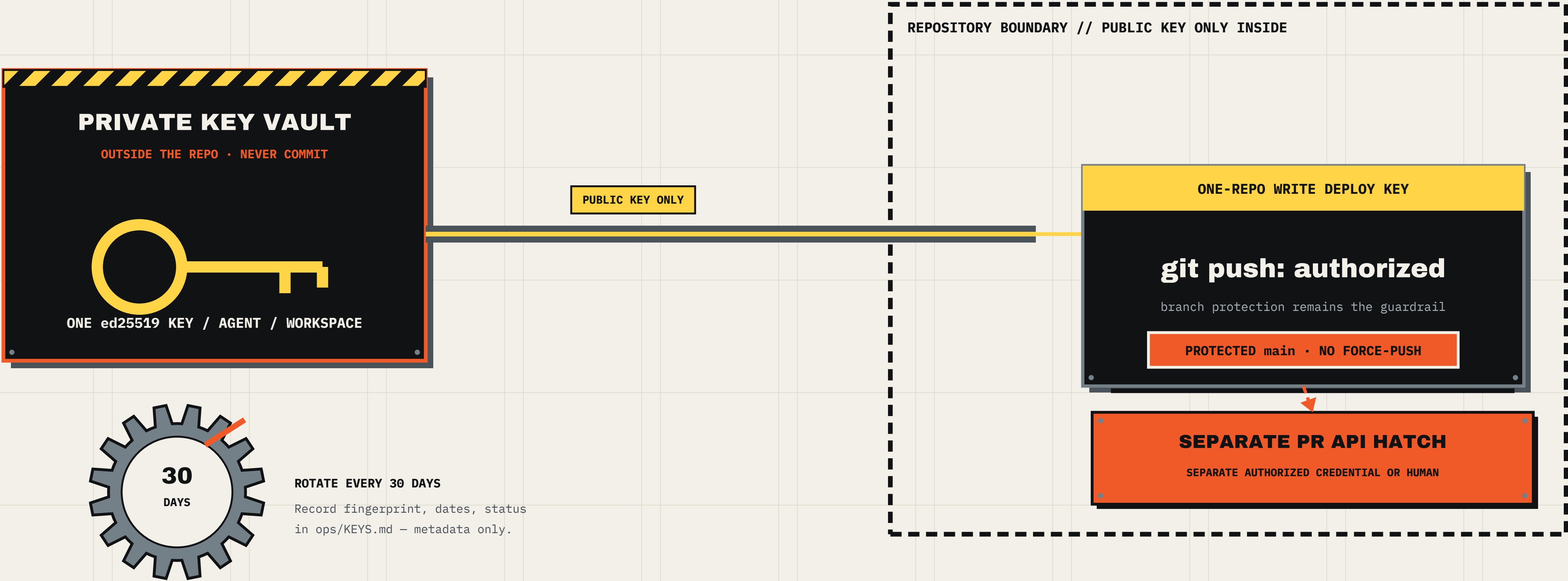
forge-lint REJECTS:

- 01 src/ without a spec link
- 02 IMPLEMENTED AC without a named test
- 03 Invalid document frontmatter
- 04 DOING task stale for >48 hours
- 05 Source change without a LEDGER append

IF ALL REQUIRED CHECKS EXIT 0 -> HUMAN REVIEW

No unearned green pass.
CI IS PROCEDURE UNTIL OUTPUT EXISTS.

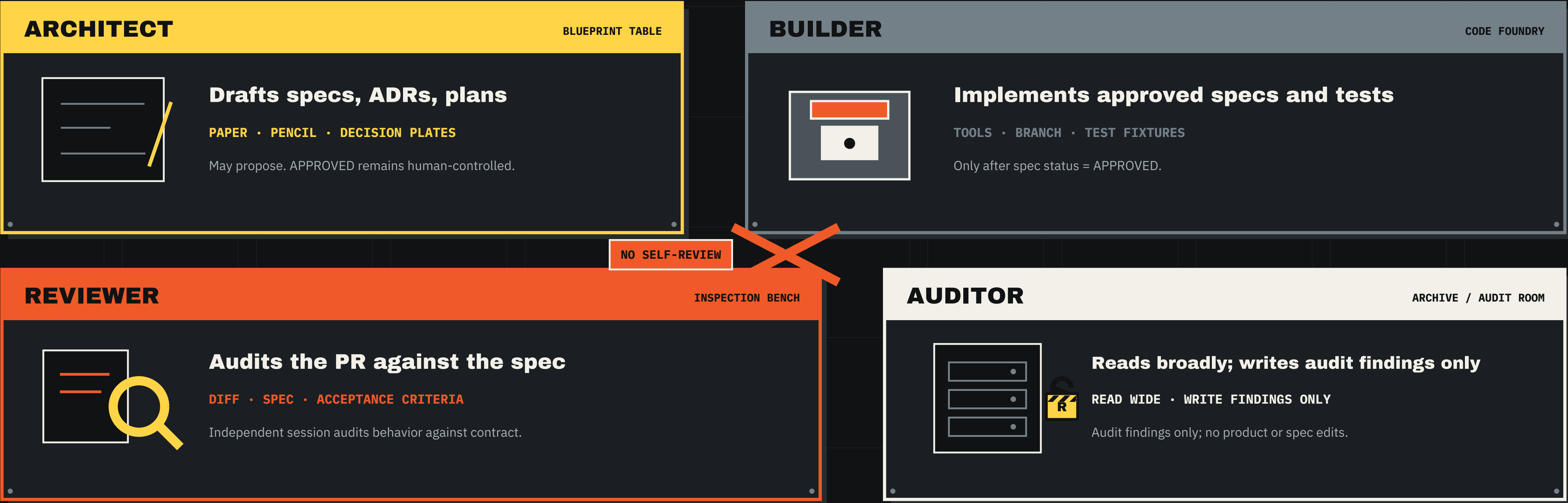
WRITE ACCESS IS A LEASE.



Deploy key = Git push. PR API access needs separate authorization.

LEAST PRIVILEGE • REVOKABLE • AUDITABLE

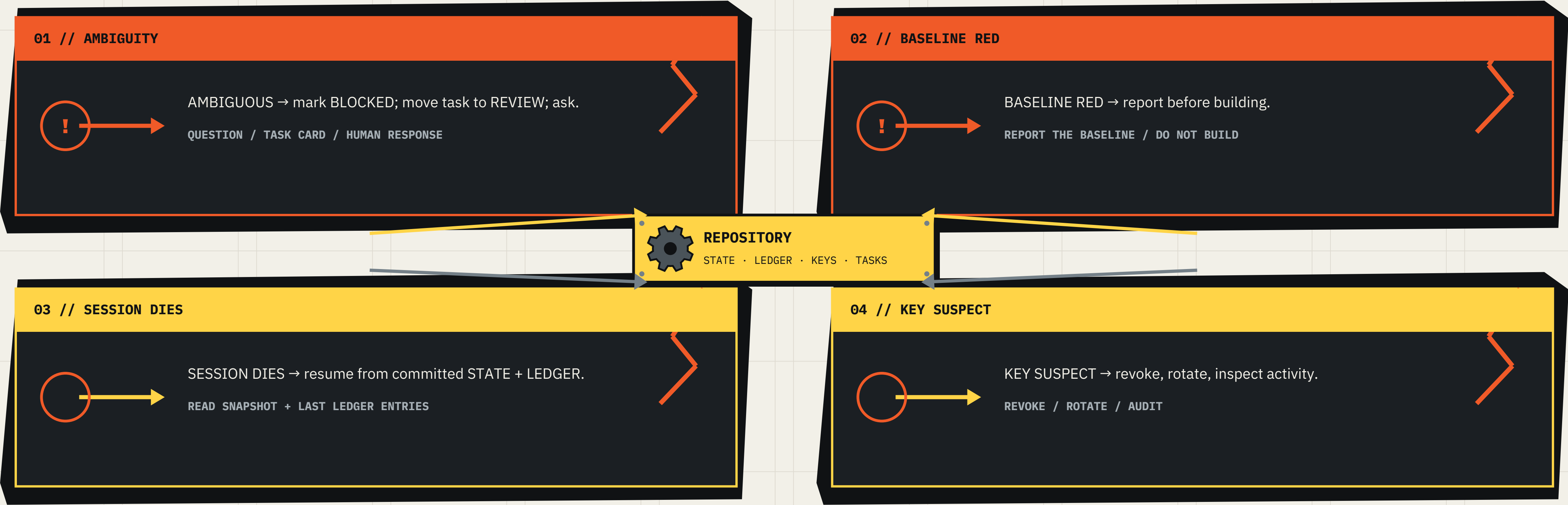
DIFFERENT HANDS. DIFFERENT POWERS.



Humans own approval, merge, and verification.

BUILDER ≠ REVIEWER • AUTHORITY DOES NOT TRANSFER BY ROLE

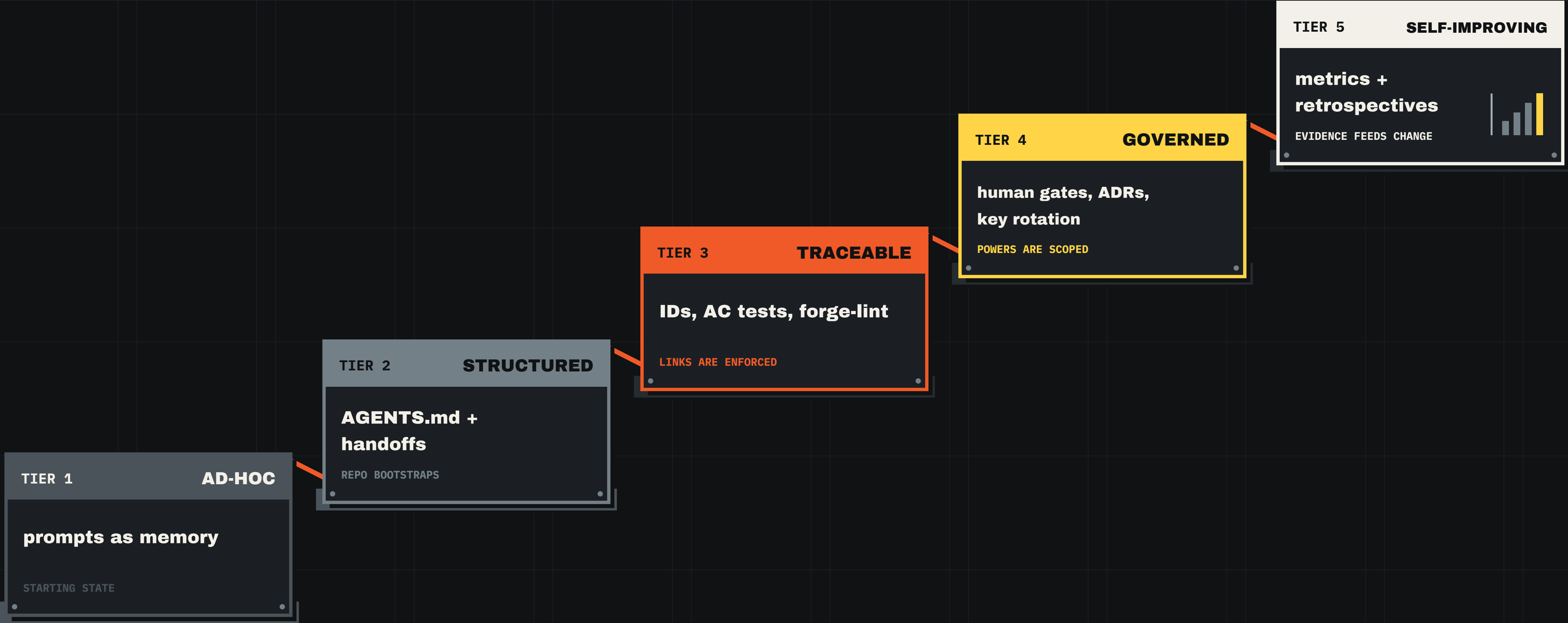
FAIL CLOSED. RECOVER IN WRITING.



Never substitute a guess for a record.
EVERY FAILURE HAS A WRITTEN REPOSITORY PATH

FIVE TIERS OF RELIABILITY.

MEASURE: spec-to-merge time · rework · blocked-task rate



START WITH ONE FILE. CONTINUE WITH ONE LINE.

FIRST SESSION // SCAFFOLD THE METHOD, NOT PRODUCT CODE

Scaffold AGENTS.md, .forge/, templates, and forge-lint. No product code.

THEN // RECURRING ONE-LINE COMMAND

“Read AGENTS.md and execute the session protocol. Priority: T-032, then T-033. Role: BUILDER.”

FIRST SESSION = METHOD ONLY

PRODUCT CODE REQUIRES APPROVED SPEC



SESSIONS END. THE SYSTEM DOES NOT.